

Cost of Production

Total Cost

- Overall basis
- Cost plus pricing
- Window of opportunity
- May opt out of lucrative deal if not handled with caution
- Budgeting, Financing
- Break even point
- Whether halt or continue
- Shutdown , Bankruptcy ?

Marginal Cost

- An operational compass to find out the rate of change in the in the total expenses
- Historical per unit performance
- Fixed cost has zero impact
- Flexible pricing
- Price war, Clearance sale, Inventory management
- Low demand period, strategic

Average Cost

- Unit Level
- Profit Margin analysis
- Scale, optimal decision
- Outsourcing
- Capacity planning
- Long term contract bidding
- Standard Retail catalogue
- Benchmark pricing
- Business viability roadmap

Output (Q)	Total Fixed Cost (TFC)	Total Variable Cost (TVC)	Total Cost (TC)	Average Fixed Cost (AFC)	Average Variable Cost (AVC)	Average Total Cost (ATC)	Marginal Cost (MC)
0	100	0	100	—	—	—	—
1	100	40	140	100.00	40.00	140.00	40.00
2	100	70	170	50.00	35.00	85.00	30.00
3	100	95	195	33.33	31.67	65.00	25.00
4	100	120	220	25.00	30.00	55.00	25.00
5	100	150	250	20.00	30.00	50.00	30.00
6	100	190	290	16.67	31.67	48.33	40.00
7	100	240	340	14.29	34.29	48.57	50.00
8	100	305	405	12.50	38.12	50.62	65.00
9	100	385	485	11.11	42.78	53.89	80.00
10	100	485	585	10.00	48.50	58.50	100.00

Cost Component	Economic Definition & Formula	Data Trends & Observations	Managerial Significance
Total Fixed Cost (TFC)	Costs that do not change with output	Constant at 100 across all output level	Represents sunk commitments (rent, salaries). Must be paid in the short run even if production stops.
Total Variable Cost (TVC)	Costs that change directly with output.	Starts at 0 , increases slowly at first up to Q=3 then increases at an accelerating rate.	Reflects expenses like raw materials and hourly labor . Directly controllable by adjusting output levels.
Total Cost (TC)	The sum of all costs incurred.	Starts at 100 and mirrors the shape of the TVC curve, shifted upward by 100.	Used to determine overall profitability against total revenue TR
Average Fixed Cost (AFC)	Fixed cost per unit of output	Continually declines from 100.00 down to 10.00 as output reaches 10.	Demonstrates the power of " spreading the overhead. " Higher production drastically dilutes fixed costs per unit.
Average Variable Cost (AVC)	Variable cost per unit of output.	U-shaped: Drops from 40.00 to a minimum of 30.00 (at Q=4,5) , then rises to 48.50.	Identifies short-run efficiency. Minimum AVC is the Shutdown Point threshold.
Average Total Cost (ATC)	Total cost per unit of output.	U-shaped: Drops steeply from 140.00 to a minimum of 48.33 (Q=6) , then rises.	Dictates per-unit profit margins. To maximize per-unit profit, market price must exceed minimum ATC.
Marginal Cost (MC)	The cost of producing one additional unit.	U-shaped: Falls from 40.00 to a minimum of 25.00 (at Q=5,6) , then rises sharply to 100.00.	The ultimate decision-making tool for expanding or contracting production.

Economic Logic & Geometric Relationships

The behaviour of these costs is entirely driven by the **Law of Diminishing Marginal Returns**:

- **Phase 1: Increasing Returns ($Q=1$ to $Q = 3$):** MC is falling (40 to 30 to 25). Teamwork, specialization, and efficient utilization of fixed machinery mean each additional unit costs *less* to make than the previous one.
- **Phase 2: Diminishing Returns ($Q = 4$ to $Q = 10$):** MC begins to rise sharply (25 to 30 to 40 to 100), Because fixed capital (TFC=100) is constant, crowding occurs as more variable inputs (labour) are added. Workers block each other, machines wear out faster, and efficiency drops.

Key Intersections to Note:

- 1. MC cuts AVC at its minimum:** At $Q=4$ and $Q=5$, AVC reaches its lowest point (30.00). MC is exactly equal to AVC (30.00) at $Q=5$.
- 2. MC cuts ATC at its minimum:** At $Q=6$, ATC reaches its lowest point (**48.33**). MC at this stage is 40.00 and rising rapidly, crossing the ATC curve right between $Q=6$ and $Q=7$.

AFC:

- An **asymptotical approach** (asymptotic approach) describes a trend where a line or value gets closer and closer to a specific target or boundary, but never actually touches or crosses it, no matter how far out you calculate it.
- Rectangular Hyperbola

ATC:

Phase I : AFC dominates: (at low level of output, AFC falls along with AVC. As both falls because of efficiency ATC also falls)

Phase II: Balance out: Minimum ATC (Upward pull of AVC and downward pull of AFC brings out the ATC to its minimum)

Phase III: AVC dominates (Upward pull of AVC because of inefficiency that completely overrides the savings in AFC, hence ATC rises.

Managerial Decisions & Window of Opportunity

Managers can look at this cost structure as three distinct operational zones depending on market demand and prices.

The "Window of Opportunity" (Optimal Capacity)

The absolute sweet spot for operational efficiency is **$Q = 5$ to $Q = 6$** .

- At **$Q = 5$** , the firm achieves the **lowest variable cost per unit** ($AVC = 30.00$).
- At **$Q = 6$** , the firm achieves the **lowest overall cost per unit** ($ATC = 48.33$).
- *Managerial Action:* This is the target production zone. It maximizes the profit buffer per unit sold.

Short-Run Pricing Strategy & Survival Rules

Managers must cross-reference their market price (P) with these calculated unit costs to make critical survival choices:

- **The Break-Even Price ($P = 48.33$):** If the market price is exactly 48.33, the firm should operate at $Q=6$). It earns zero economic profit but covers all explicit and implicit costs.
- **The Profit Zone ($P > 48.33$):** If market price is above 48.33 (e.g. $P = 65$), the firm should expand production. It should look at the **($MR = MC$)** rule. If price is 65, it should produce exactly **($Q = 8$)** (where $MC = 65$) to maximize total profit.

Short-Run Pricing Strategy & Survival Rules (continued...)

- **The Loss-Minimization Zone ($30.00 < P < 48.33$):** If price drops to 40.00, the firm cannot cover total costs. However, because 40.00 is *above* the minimum AVC of 30.00, the firm **should continue operating** in the short run. By producing, it fully covers its variable costs and has a little left over to pay down a portion of its fixed costs. Shutting down would result in losing the full 100 fixed cost.
- **The Shutdown Point ($P < 30.00$):** If market price drops below 30.00, the firm cannot even afford to buy raw materials or pay immediate labor for production. **The manager must shut down operations immediately** to cap losses at the fixed 100.

The Core Economic Principle (The "Grade Point Average" Logic)

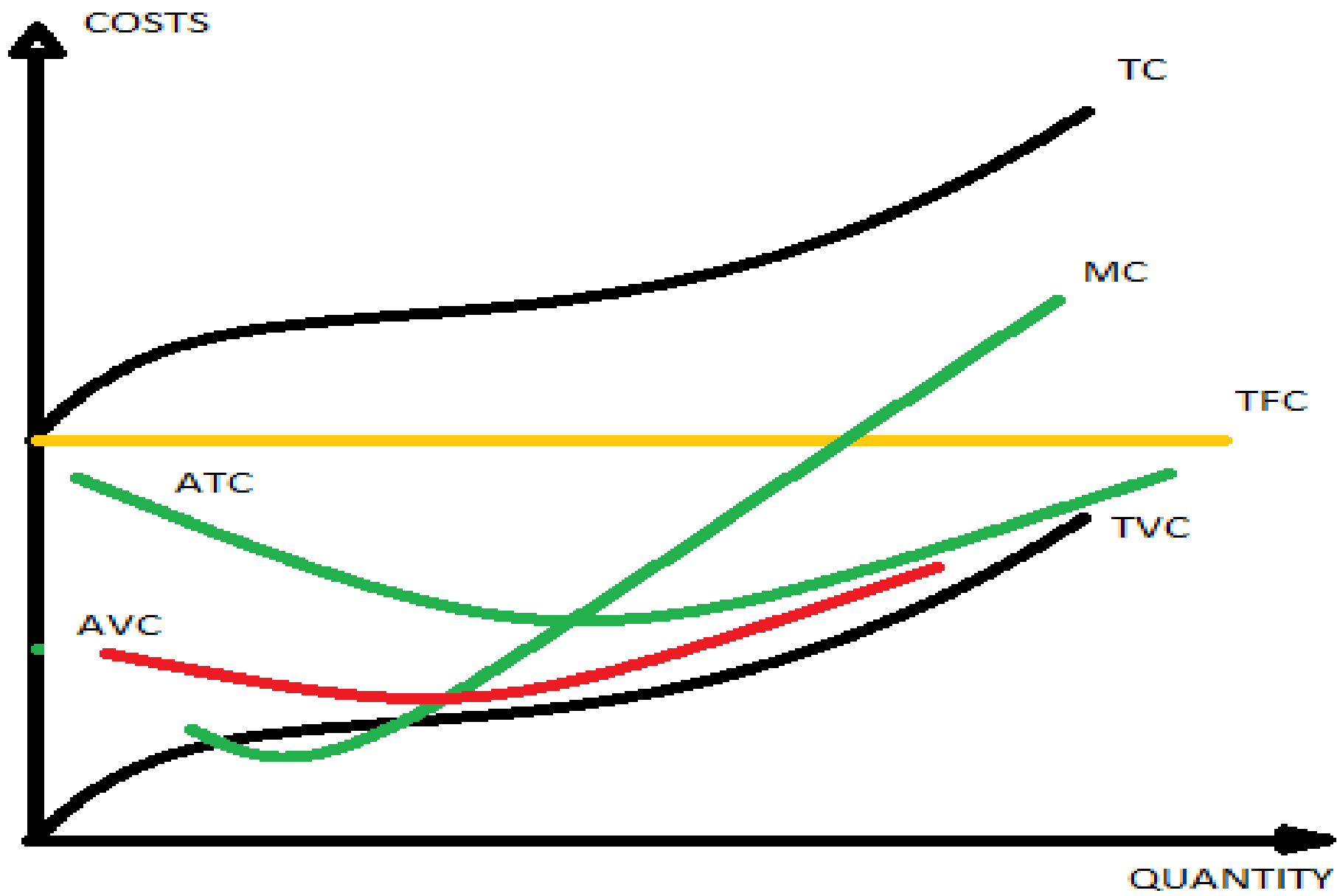
To understand the relationship intuitively, think of **AVC** as your cumulative GPA and **MC** as the grade you get in your very next class.

- **When MC is below AVC (Pulling the Average Down):** If your current cumulative GPA (AVC) is 3.5, and you get a 2.0 (MC) in your next class, your cumulative average *must* fall. Economically, if producing the next unit costs less than the average cost of all previous units, it drags the average variable cost downward.
- **When MC is above AVC (Pulling the Average Up):** If your GPA (AVC) is 3.5, and you pull a 4.0 (MC) in your next class, your cumulative average *must* rise. Economically, if the next unit costs more to produce than the existing baseline average, it drags the average variable cost upward.
- **The Intersection at the Minimum:** If a number pulls the average down when it is below it, and pulls the average up when it is above it, the average can *only* stop falling and start rising at the exact moment the next addition is **perfectly equal** to the average. This transition point is, by definition, the absolute lowest point (the minimum) of the average. Therefore, **AVC stops falling and hits its minimum exactly when $MC = AVC$.**

The Physical Cause in the Factory (Why Costs Change Shape)

In the real world, the U-shape of these curves—and their crossover point—is driven entirely by the **Law of Diminishing Marginal Returns**.

1. **Increasing Efficiency ($MC < AVC$):** The first few workers you hire specialize. They become highly productive, meaning the additional cost to make each new product (**MC**) drops rapidly. Because these products are so cheap to make, they actively pull down the average cost (**AVC**).
2. **The Bottleneck (MC turns upward, but is still below AVC):** Eventually, Workers start waiting and Diminishing returns set in. The cost of making the product (**MC**) begins to rise. However, even though MC is rising, **it is still lower than the AVC** , so the running average continues to fall.
3. **The Crossover ($MC = AVC$ at Minimum AVC):** As you keep adding workers, MC keeps climbing sharply. At a specific output level, the cost of making that next extra product rises to match the exact running average cost. Because the next unit no longer costs less than the average, **AVC stops falling**. It has hit rock bottom (its minimum).
4. **Inefficiency Explodes ($MC > AVC$):** If you hire even more workers, the shop floor is chaotic. The cost of making one more product(**MC**) goes up and is now higher than the average. This extra expense drags the running **AVC** upward, creating the right side of the U-shape.



The Multi-Scale Cost Schedule (SRATC₁, SRATC₂, SRATC₃)

Output (Q)	SRATC₁ (TFC=100)	SRATC₂ (TFC=250)	SRATC₃ (TFC=500)	Optimal Plant Choice (LRATC)
2	85.00	152.50	274.00	SRATC₁ (85.00)
4	55.00	85.50	146.25	SRATC₁ (55.00)
6	48.33 (Min)	61.67	102.17	SRATC₁ (48.33)
8	50.62	50.25	81.50	SRATC₂ (50.25)
10	58.50	44.50	69.60	SRATC₂ (44.50)
12	71.25	42.00 (Min)	61.67	SRATC₂ (42.00)
14	89.20	44.07	53.46	SRATC₂ (44.07)
16	112.50	50.12	46.00	SRATC₃ (46.00)
19	160.00	64.31	44.84 (Min)	SRATC₃ (44.84)
22	225.00	88.50	48.05	SRATC₃ (48.05)

LRATC curve is just a collection of those individual short-run tables. When management looks at the long-run table, they are looking at different short-run tables side-by-side so they can pick the absolute lowest number in each row to form the **LRATC envelope**.

Visualizing the Envelope Effect

The **Long-Run Average Total Cost (LRATC)** curve acts as an **envelope** because it wraps around the bottom of all short-run choices. In the long run, a business will never choose to operate on a high-cost curve if they can switch plant sizes to achieve a lower per-unit cost.

- **From $Q = 1$ to 7:** The manager stays with **SRATC₁**.
- **At $Q = 8$:** The crossover happens. The manager breaks ground on a Medium Plant because **SRATC₂ (50.25)** becomes cheaper than continuing to overstretch the small plant on **SRATC₁ (50.62)**.
- **At $Q = 15$:** The manager transitions to **SRATC₃** because massive volume allows the big facility to dilute its \$500 overhead.

Stage 1: Economies of Scale (The Downward-Sloping Path)

- **What is happening:** As the firm scales up from a small plant to a medium plant, the absolute lowest possible per-unit cost drops from **48.33** (at $Q=6$) down to **42.00** (at $Q=12$).
- **Economic Logic:** This occurs because larger operational structures unlock **technological efficiencies and specialization**. In a larger factory, workers can focus on single tasks instead of multitasking, bulk raw materials can be purchased at steep discounts, and specialized heavy machinery can be fully utilized instead of sitting idle.

Stage 2: Constant Returns to Scale (The Flat Bottom / Transition)

- **What is happening:** Between $Q = 12$ and $Q = 19$, per-unit costs reach their optimal floor, leveling off in the **\$42.00 to \$44.00 range**.
- **Economic Logic:** The firm has reached its **Minimum Efficient Scale (MES)**. At this point, the engineering gains from building an even larger plant ($SRATC_3$) are almost perfectly balanced out by the expanding overhead costs (\$500). Doubling inputs results in exactly double the output, keeping per-unit costs stable.

Stage 3: Diseconomies of Scale (The Upward-Sloping Path)

- **What is happening:** If the firm expands blindly past **$Q = 19$** , the LRATC begins ticking upward again, rising to **48.05** at $Q=22$ and climbing higher.
- **Economic Logic:** This is caused by **managerial bottlenecks and bureaucratic friction**. When an enterprise becomes too vast, communication channels break down, management layers multiply, and it takes longer to make simple decisions.

Metric / Attribute	Short-Run Average Total Cost (SRATC)	Long-Run Average Total Cost (LRATC)
Input Flexibility	At least one input (e.g., capital, factory size) is fixed .	All inputs (labor, capital, land) are variable .
Cost Composition	Includes both Fixed Costs and Variable Costs .	Includes Variable Costs only (no fixed costs).
Curve Determinant	Shaped by the Law of Diminishing Marginal Returns .	Shaped by Economies and Diseconomies of Scale .
Relative Position	Always equal to or higher than the LRATC curve.	Always equal to or lower than any given SRATC curve.

Algebraic Condition	Type of Scale Return	Long-Run Economic Impact
$\alpha + \beta > 1$	Increasing Returns to Scale (IRS)	This creates economies of scale , forcing the LRATC curve downward .
$\alpha + \beta = 1$	Constant Returns to Scale (CRS)	Output grows in exact LRATC remains perfectly flat.
$\alpha + \beta < 1$	Decreasing Returns to Scale (DRS)	This creates diseconomies of scale , tilting the LRATC curve upward .

Region of LRATC	Visual Behavior of LRATC	Tangency Point on the SRATC Curve
Economies of Scale	Sloping Downward	Left of the SRATC minimum point
Minimum Efficient Scale	Perfectly Flat (Minimum)	Exact Minimum of the SRATC curve
Diseconomies of Scale	Sloping Upward	Right of the SRATC minimum point

